

# Frontier AI for Actuaries.

AI helps actuaries perform complex, multi-step actuarial tasks with every step documented and reproducible.

## THE WORKBENCH

### One collaborative workspace for pricing, reserving, and filing

For all actuaries at admitted and specialty carriers, MGAs, and reinsurers.



# Actuarial work is not one workflow. It is a web of intricate, connected tasks.

The more steps an actuary delegates to AI, the less certain they can be of the output.

Tesora enables full tracing, documentation, and audit, so the actuary remains in control.



**Philo Bishay** ✓ • 1st

Cofounder & Head of Product @ Tesora | Actuarial | Wharton MBA

2w • 🔄

Every actuary dreads one question, "What do actuaries do?" I am happy to say we are no closer to an answer, but if you want to avoid further questions, send this:

## What Actuaries Deliver

Two questions. All decisions.  
What do we charge?  
What do we already owe?  
Actuaries own both.

Legend:  
■ Shared Foundation  
■ Admitted (Filed Paper)  
■ E&S / Non-Admitted  
■ Pricing Merge  
■ Reserving  
→ Primary Flow (8)  
→ Learning Feedback (2)

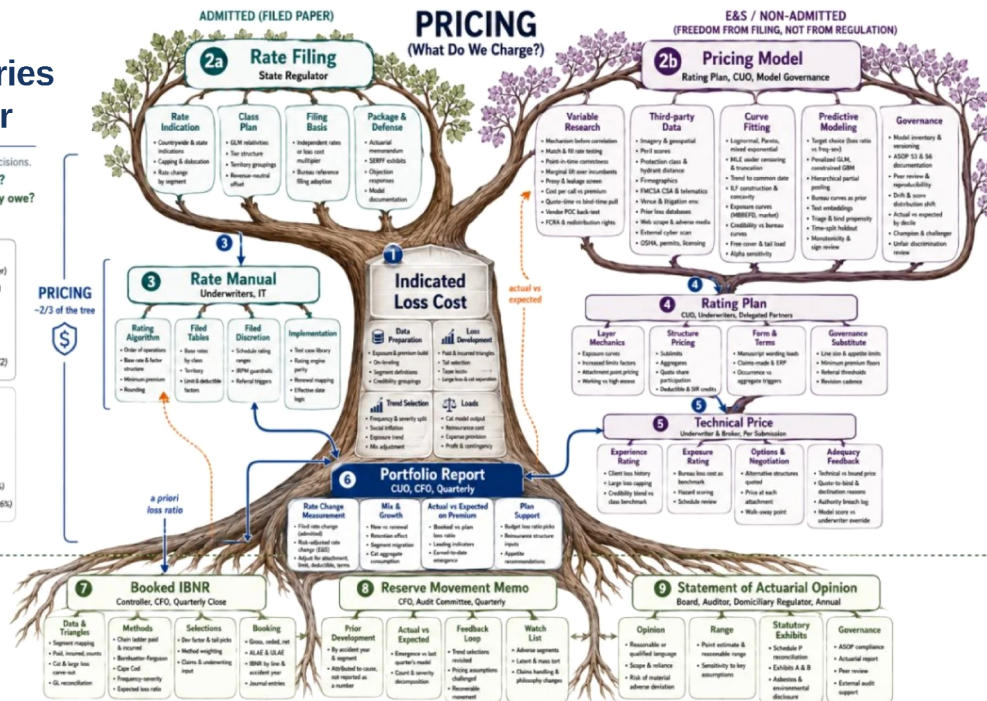
AT A GLANCE  
10 Major Deliverables  
41 Branches  
163 Leaves  
121 Pricing Leaves (24%)  
42 Reserving Leaves (26%)



RESERVING  
~1/3 of the tree



Note: Admitted and non-admitted describe the paper, not the company, since filing status is decided state by state and risk by risk, and a single carrier group routinely runs both lines at once.  
Freedom of rate and form is freedom from rate filing, not freedom from rate regulation: surplus lines carriers remain subject to unfair discrimination statutes, to state adoption of the NAIC Act model bulletin, and to their domiciliary regulator, reinsurers, and auditors.



... Over 150 workflows are core to the actuarial function



■ 121 pricing ■ 42 reserving

one mark per workflow

Order of operations

Base rate and factor structure

Limit and deductible factors

Schedule rating ranges

IRPM guardrails

Rating engine parity

- Chain ladder, paid and incurred
- Bornhuetter-Ferguson
- Cape Cod
- Development factor and tail picks
- Cat and large loss carve-out
- IBNR by line and accident year

Ingest

Triangles

Trend

Indication

Filing

One weak handoff and everything downstream is unusable.

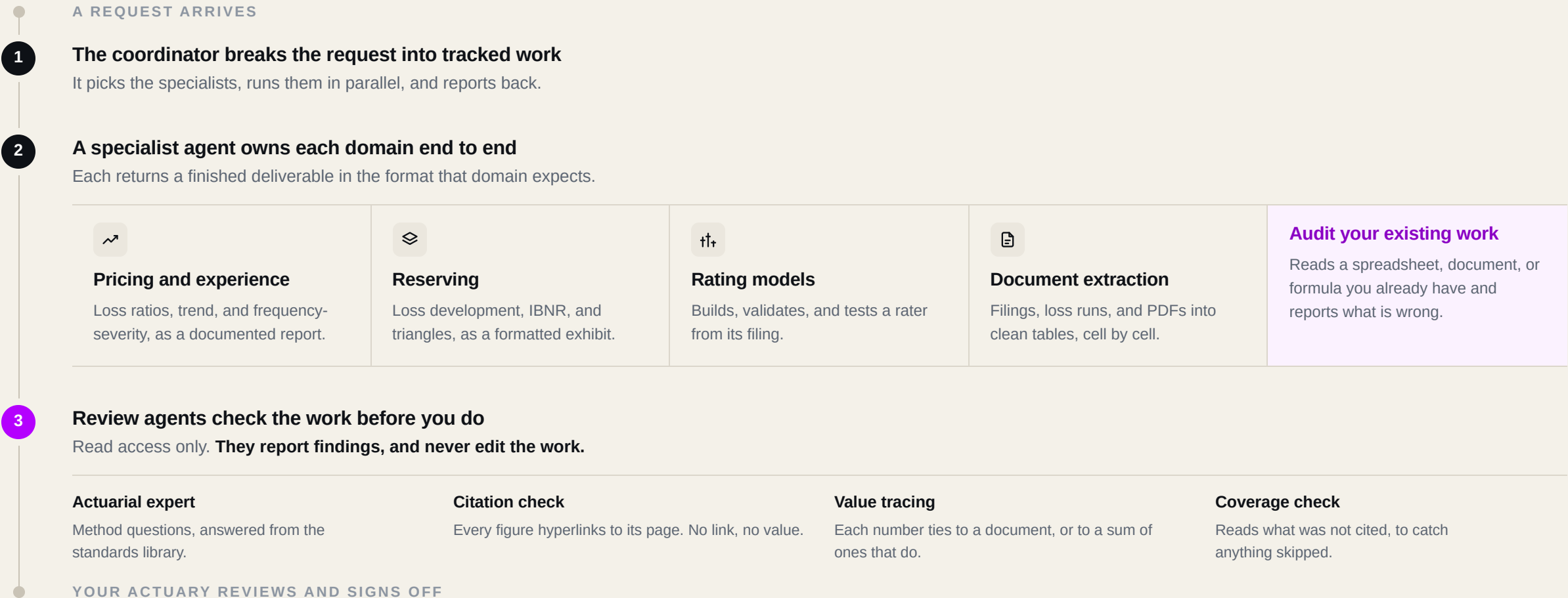
Ask a model to do all of it in one shot and you get a confident number nobody can trace.





# No single agent does the whole job, and **none of it reaches you unchecked.**

We watched how actuaries actually do the work, mapped it into sub-agents, and wrote evals for each one, with audit agents sitting behind them. In a domain this intricate, that is what delivers a business outcome without a hallucinated number or a botched handoff between steps.





# Rate changes in **days**, not months.

Source a rater from a SERFF filing, an Excel file, or a script. Tesora rebuilds it, tests it, and deploys it whenever underwriting is ready.

## BEFORE: BY HAND

# 4-6 months

From a requested change to a filed, live rater, and where the months actually go

- |    |                                                                                       |       |
|----|---------------------------------------------------------------------------------------|-------|
| 01 | <b>Spec the change</b><br>Actuary writes up the revised rates and factors             | DAYS  |
| 02 | <b>IT ticket, then the queue</b><br>The change waits for developer capacity           | WEEKS |
| 03 | <b>Re-code the rater</b><br>Engineering reimplements the logic by hand                | WEEKS |
| 04 | <b>Rebuild exhibits, 50 states</b><br>Each state reformatted by hand for SERFF        | WEEKS |
| 05 | <b>Manual QA</b><br>No test suite, so regressions are caught by eye                   | WEEKS |
| 06 | <b>Compliance sign-off</b><br>Actuarial and legal review the filing                   | DAYS  |
| 07 | <b>File and answer objections</b><br>Back and forth with each department of insurance | WEEKS |
| 08 | <b>Release window</b><br>Deploy only in the next scheduled window                     | WEEKS |

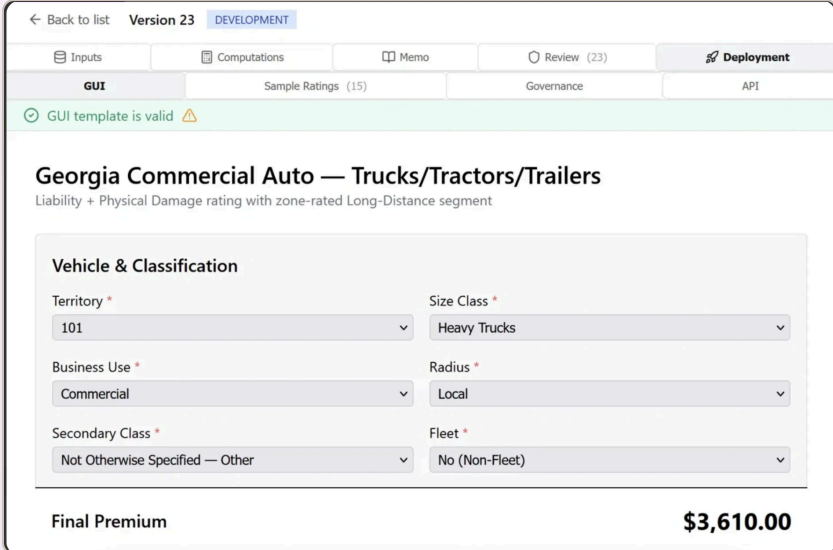
## AFTER: Tesora

# 1-2 days

Deploy to a GUI, an API, or back into Excel.

**You give:** a plain-language change and the source rater, from SERFF, Excel, or a script.

**You get:** a tested rater with a working GUI, sample ratings, a versioned API, or an Excel file your team can open.







# The triangle is squared before you sit down.

A quarterly close moves between actuarial, IT, and finance, and most of the calendar goes to assembling the data rather than judging it.

## BEFORE: BY HAND

~2 weeks

From close to a booked IBNR, and who is waiting on whom

### 01 Request the data

IT

Actuarial files a ticket for the quarter's extract

### 02 Wait on the extract

DAYS

The pull sits behind other reporting work

### 03 Reconcile to the ledger

FINANCE

Paid and incurred tied back to finance, by hand

### 04 Rebuild the triangles

DAYS

Segment mapping and large-loss carve-outs redone each quarter

### 05 Select factors and methods

ACTUARIAL

Chain ladder, Bornhuetter-Ferguson, Cape Cod, tail

### 06 Rebuild the exhibits

DAYS

Roll-forward and movement memo reformatted by hand

### 07 Peer review

DAYS

A second actuary retraces the work from scratch

### 08 Book and explain

DAYS

Answer the CFO, the auditor, and the board

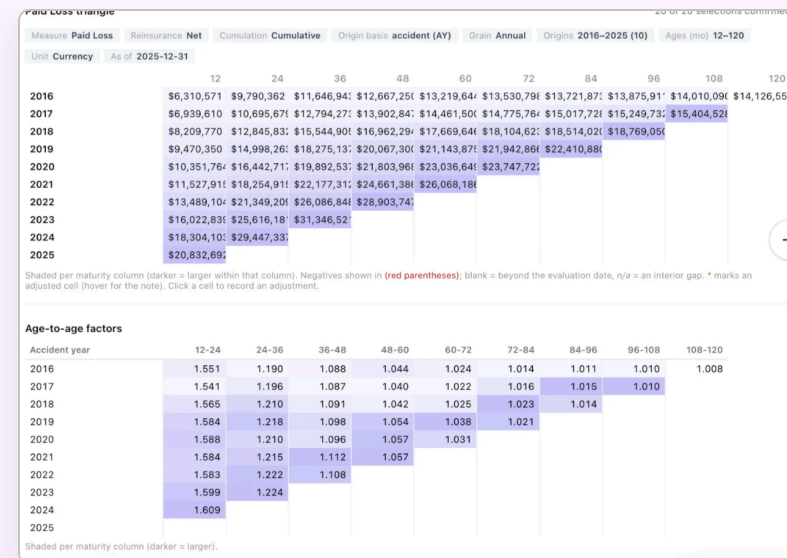
## AFTER: Tesora

3-4 hours

Triangles squared on ingest, so the quarter starts at the selection

**You give:** the loss run, as it comes out of the system.

**You get:** reconciled triangles, method comparisons, and an IBNR with every value traced to source.





# Rebuild what a competitor charges, then **price your book through it.**

A filing gives you tables, rules, and inputs rather than a working model, and the order of operations is rarely spelled out. Turning that into something you can run your own risks through is the work.

## BEFORE: REBUILT BY HAND FROM SERFF

**~1 week**

From a competitor filing to a usable read, and where it goes wrong

- |                                                 |                |
|-------------------------------------------------|----------------|
| <b>01 Pull the filings</b>                      | <b>HOURS</b>   |
| An analyst searches SERFF, state by state       |                |
| <b>02 Read the rate manual</b>                  | <b>DAYS</b>    |
| Rules, factors, and order of operations, by eye |                |
| <b>03 Rekey the rate tables</b>                 | <b>DAYS</b>    |
| Hundreds of tables retyped into a spreadsheet   |                |
| <b>04 Rebuild the algorithm</b>                 | <b>DAYS</b>    |
| The logic is inferred, then argued over         |                |
| <b>05 Price a sample</b>                        | <b>HOURS</b>   |
| A dozen test risks, never the real book         |                |
| <b>06 Compare by hand</b>                       | <b>DAYS</b>    |
| Deltas built in a one-off workbook nobody owns  |                |
| <b>07 Explain the gaps</b>                      | <b>DAYS</b>    |
| Each difference traced back to a single factor  |                |
| <b>08 The read goes stale</b>                   | <b>ONGOING</b> |
| They filed again while you were rebuilding      |                |

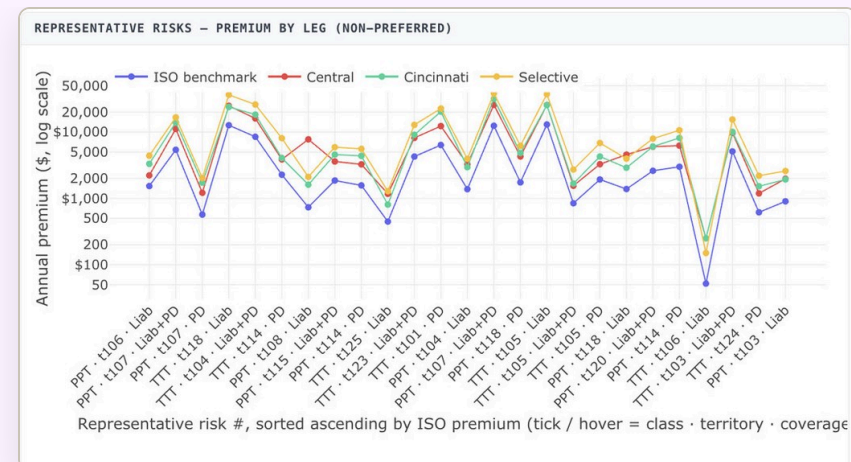
## AFTER: Tesora

**10-15 minutes**

An executable competitor rater, and your whole book priced through it

**You give:** a competitor, a state, and a line. Or the SERFF filing itself.

**You get:** their rating logic running against your risks, every factor cited to the page it came from.



Built from public rate filings, and any additional data sources you provide.



# The full context is baked into the work.

Every session holds the data it used, the assumptions behind it, and the reasoning that got there. Reopen it years later and nothing has to be reconstructed, because the context was never separate from the answer.

The screenshot displays the Tesora platform interface. On the left is a sidebar with navigation options: Home, Work Tracking, ARTIFACTS (Folders, Reports, Raters, Routines), APPS (Compute St..., Underwriting, Rater Recon..., Competitive ..., GLM Studio, Triangle Bull..., All Apps, Support, Settings), and a user profile for 'vivek'. The main area is divided into three panels. The left panel shows a 'Rate Rule Form' with a table of metrics:

| Metric                          | Value      |
|---------------------------------|------------|
| Profit & Contingencies          | 7.0%       |
| Total expense & profit load     | 47.0%      |
| Expected Loss & LAE Ratio (ELR) | 53.0%      |
| Projected Ultimate Loss Ratio   | 54.7%      |
| Adequacy margin (ULR - ELR)     | 1.7% short |

Below the table, a 'Read' section states: 'the thin spot is the 47% load (30 pts of it commission), and the book runs -1.7 pts past its permissible loss ratio — a modest inadequacy with no cushion.' A 'Caveat' section notes: 'ILFs are relativities, not dollar prices; no competitor base rates or retention/deductible tables exist in the folder. Preliminary analyst work product, subject to credentialed-actuary review.' A note mentions a CSV download: 'CIC\_vs\_Peers\_ILF\_Competitive\_Benchmark.csv'. The right panel shows a document titled 'bernegger-2024-integrated-experience-exposure-rating.pdf' with a 'Download' button. Below the document title is a 'SUMMARY' section with the text: 'No summary available. The ingestion workflow either hasn't finished or didn't produce one for this document.'

## Implementation

The models are wired into your loss runs, policy systems, and ledger, running inside your own environment.

## Governance

Author, reviewer, and approver are separate identities, and the trail survives every promotion. Finance and IT find what they need without waiting on an actuary.

## Compliance

Built to ASOP from the start, and SOX-aligned for carriers with public-company parents. Same input, same answer, every time.

## Continuity

When an actuary leaves, their assumptions, sources, and reasoning stay in the session. The next person picks up the thread.

Every request from the platform pings our chief actuary and CTO. Support is 24/7, agents and people together, so you are never left with a bot. A real person answers within two to three minutes, and we have had no unplanned downtime to date, which is **model routing** doing its job.



# What's the best way to **implement AI**?

Your implementation of AI today will dictate its value, translatability, and reproducibility going forward.

|                                  | <b>Actuarial consultants</b><br>Milliman, WTW, Oliver Wyman          | <b>Your team with AI</b><br>In-house, on ChatGPT or Claude              | <b>Actuarial software</b><br>hyperexponential, Earnix, Akur8             | <b>Tesora</b><br>Frontier models, in your format                                    |
|----------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Turnaround</b>                | <input type="radio"/> Weeks to months, in their queue                | <input checked="" type="radio"/> Quick drafts, still checked by hand    | <input checked="" type="radio"/> Fast and repeatable once it is live     | <input checked="" type="radio"/> Minutes to days, in your workflow                  |
| <b>Setup &amp; fit</b>           | <input checked="" type="radio"/> A fresh engagement scoped each time | <input checked="" type="radio"/> No one knows your book better          | <input checked="" type="radio"/> Heavy implementation on their platform  | <input checked="" type="radio"/> 60-day build in your VPC, on your own data         |
| <b>Your raters &amp; filings</b> | <input type="radio"/> Generic deliverables in their template         | <input checked="" type="radio"/> Your format, rebuilt by hand each time | <input checked="" type="radio"/> Rebuilt inside their system, their way  | <input checked="" type="radio"/> Your raters, versioned and ready to deploy         |
| <b>Source &amp; audit</b>        | <input checked="" type="radio"/> Independent, credible workpapers    | <input type="radio"/> No source to trace, and the answer drifts         | <input checked="" type="radio"/> Logged inside the platform              | <input checked="" type="radio"/> Cell-level citations, append-only log              |
| <b>Scope</b>                     | <input checked="" type="radio"/> Whatever you scope and pay for      | <input checked="" type="radio"/> One step at a time, gaps in between    | <input checked="" type="radio"/> Pricing or reserving, rarely filing too | <input checked="" type="radio"/> Pricing, reserving, filing, and close in one place |

☒ Full capability   ☐ Gap

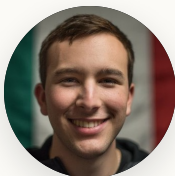


A team built at the intersection of frontier AI research and decades of actuarial experience, **the two disciplines this problem demands.**

**Vivek Rao**

CEO

Former private-equity investor and McKinsey consultant focused on P&C insurance.

**Federico Reyes**

CTO

Stanford AI researcher. Graph neural nets and relational data at Kumo AI, acquired by Nvidia.

**Philo Bishay**

CPO &amp; CHIEF ACTUARY

Former chief actuary at W. R. Berkley E&S. Capital modeler at Gen Re; built and sold a brokerage.

**Michael Palmich**

HEAD OF SALES

25+ years in the actuarial software space, most recently at Akur8 and Milliman.

**Eric Lam**

FOUNDING ENGINEER

MIT for CS, physics, and math, with **three actuarial exams** done.

## ADVISORY BOARD

**Ethan Genteman**

FCAS, MAAA

Chief Actuary, Palomar (NYSE: PLMR)

**Mario DiCaro**

FCAS, CERA

Head of Capital Modeling, Tokio Marine HCC

**Joanne Chen**

General Partner, Foundation Capital

BACKED BY



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# Above all, we want to help you **win**.

We are a thought partner first. This technology is powerful, and it only helps once it is opinionated: built with a real point of view on how the work gets done, how it gets checked, and how it gets defended when a regulator asks.

We are here to push the frontier forward with the right partners, not to sell you a license. A law firm does the legal work rather than selling software to lawyers, and we hold ourselves to that standard: **we take ownership of the outcome, not just the tooling**. Every deployment starts with a proof of concept anchored on your own success metrics, and we work with your team to find where the workbench sets the shared standard first.

Let's set up a call

[sales@tesora.ai](mailto:sales@tesora.ai)

## DE-RISKED BY DESIGN

- ✓ It runs in your own environment, in a dedicated tenant. No cross-tenant inference.
- ✓ Your rate plans, filings, and books are never used to train shared models.
- ✓ AES-256 at rest, TLS 1.3 in transit, and your own SSO.
- ✓ We clear your security review before any work touches a filing.



SOC 2 Type II. Independent auditors review us annually, and our trust center shows the continuous testing live. SOX-aligned against both actuarial and IT standards.

# Appendix

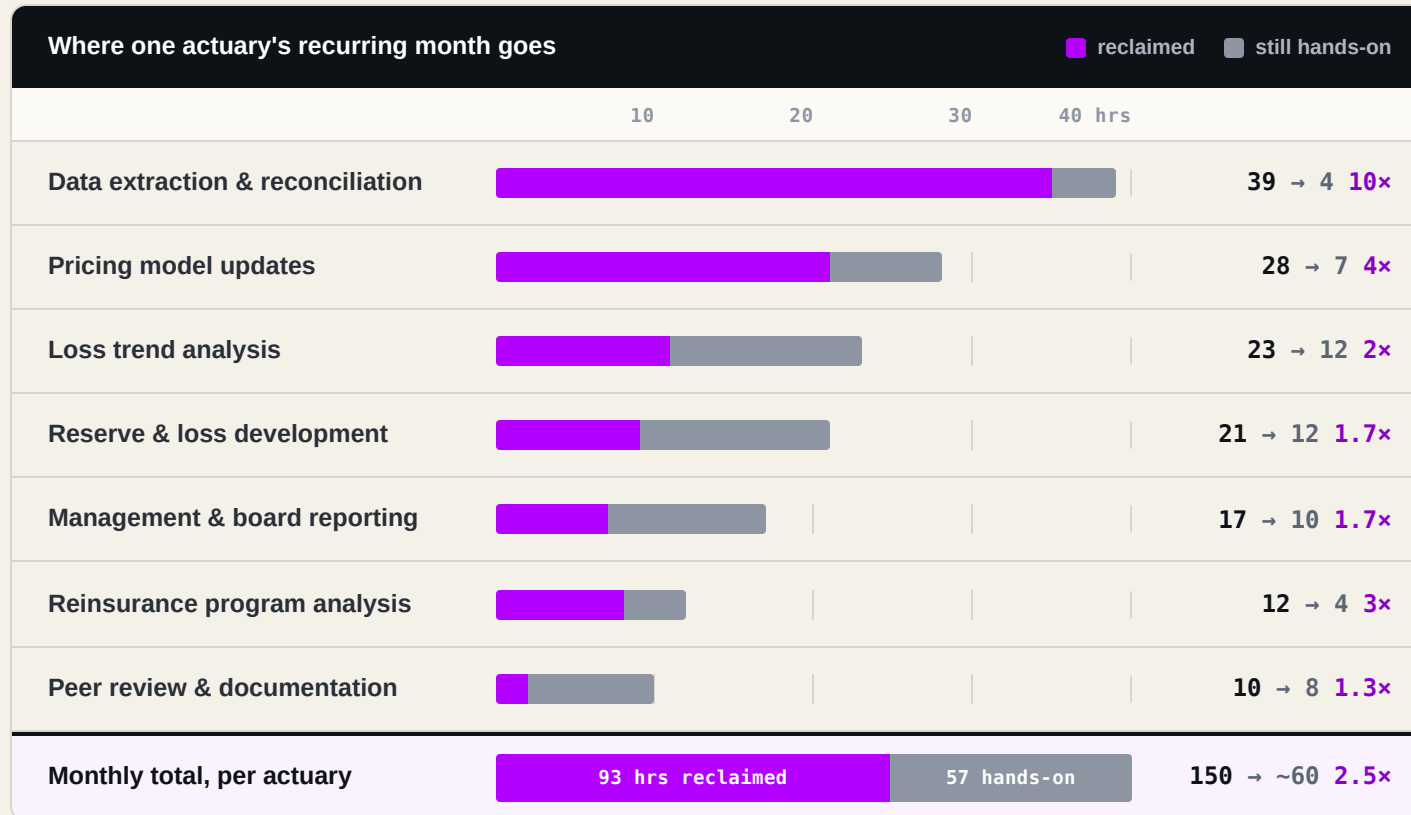




## Per actuary, the recurring month goes from **150 hours to about 60.**

Most of the month disappears into recurring work: the data pulls, the reconciliations, the model updates.

Tesora compresses that mechanical layer, so the hours come back as judgment rather than assembly.



### ~90 hours back per actuary, per month

A 2.5× blended throughput gain across the recurring month, with the same team and the same sign-off.

### Smarter, not harder

Time freed for strategy and selections rather than assembly.

### Unlock your actuaries

More products, more territories, and more actuarial capability from the same team.

*Hours are per actuary and reflect a typical ~150 hour recurring workload within a roughly 160 hour month, with per-workflow gains from Tesora's engagement framework. Figures are modeled and confirmed against measured results during a build.*



# We run on top of Claude, and we are model agnostic.

The model is the engine, not the product. What an actuary needs around it is infrastructure: the data wired in, the workflow decided, the cost controlled, and the record kept. That is the part we build, and we build it for you.

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## We guide the model

A raw model answers anything, confidently. Ours is held to an actuarial method: fit frequency and severity separately, test against your own history, and stop to ask before selecting.

---

## We wire the infrastructure

Your loss runs, policy systems, filings, and ledger are connected once and stay connected, so every study reads the same reconciled book. A chat window starts empty every time.

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## The workflow is opinionated

Reserving and pricing have a right order of operations, and we encode it. Chain ladder, Bornhuetter-Ferguson, and Cape Cod run side by side, so the selection stays your actuary's call.

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## We manage the cost

Running a real book through a frontier model naively is slow and expensive. Extraction runs on a small model, and the judgment calls go to the strongest one.

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## We use the best models. You keep the value.

Every firm is working out its AI spend internally right now. Our team of AI researchers and former investors knows what these systems cost, and how to optimize model spend per business outcome.



### Right model, right step

Each step goes to the model that gets it right for the least spend. Extraction runs small, judgment runs strong, and none of that routing is your problem to manage.



### Our incentive is the outcome, not the token count

A model vendor is never going to tell you to buy less of what it sells. Our platform has to deliver the result, so keeping model spend down is work we have to solve for ourselves rather than work we are merely free to do. **You are never left at the mercy of a rogue agent burning through tokens.**



### Latest models, always up

Your workflows, data wiring, and audit trail live with us, so the strongest model arrives without a migration or a new vendor. A system built in-house on one provider stops when that provider does. We route across them, so **servicing an outage is our problem rather than your stopped quarter.**



### Context mapping

We map your data, formats, and quirks against the work they feed, so we know intimately which model fits where. That map deepens the longer we work with your team, and it carries across every upgrade.



### Audits matter more, not less

Better models take on more complex work, which raises rather than lowers the bar on proving it. Every value stays cited and reproducible, and **the advantage compounds for whoever moves first.**